Global Memory Tech

Weekly theme: Shareholder return upside, 4Q guidance preview, resilient spot/export

Industry Overview

Expect additional dividends/buybacks from Hynix/Samsung

We see potential catalysts in additional shareholder returns from SK Hynix (a special year-end cash dividend following its recent W40tn buyback) and Samsung Electronics (a new buyback post-3Q cash dividend of W30tn). Both could address these returns at their respective earnings calls on 28-29 Oct. We expect Korean memory chipmakers to step up dividends and buybacks, given exceptionally low P/E ratios (following severe share-price corrections) and strong growth in FCF (despite concerns about a downturn). As of 4Q26, annualized FCF already exceeds W200tn/W300tn for Hynix/Samsung, implying potential annual shareholder returns of W100tn+/W150tn+. We retain our assumed buyback:dividend split of 60:40 for Hynix (holdco SK Square owns only a 21% stake), vs 40:60 for Samsung (far more retail shareholders, incl. the Lee family, than Hynix). Neither company has announced new dividend or buyback plans, but management at both may view current share prices as too low relative to fundamentals. We will host a large-group webcast with TrendForce on 10 Nov to reassess the memory super-cycle and AI boom.

Bullish guidance likely; cycle differs from 2018 peak

We expect bullish guidance at next week's earnings calls. **Hynix:** The successful ramp-up of 12-hi HBM4 (50%+ of total HBM sales by end-4Q26); additional 8-hi HBM4 production if NVIDIA pays a price premium per bit of ~10% (or if margins are better than for 12-hi); further conventional DRAM ASP upside in 1H/2H27 vs the 4Q26 peak, given the ongoing chip supply shortage (vs AI demand), completion of phase 1 of the new Yongin fab in Korea in Feb'27, near-complete migration to the 1c node at Korea's conventional DRAM fab, and progress in preparing HBM4e samples on the 1c node. **Samsung:** HBM market-share gains; quarterly contract price upside even under LTAs, and strong growth in long-term orders, including agreements for chip deliveries in 2029-30 at high prices; accelerated ramp-up at Pyeongtaek P4 and construction of P5; HBM-focused migration to the 1c node (minimal migration for LPDDR5 even in mid-2027); limited support for de-spec orders (Samsung unlikely to expand 8-hi HBM capacity but to more actively expand 12-hi HBM4, followed by HBM4e in 2027); and memory chip supply to remain broadly tight relative to demand even in 2028 (current channel inventories generally lean, vs exceptionally high in 4Q18 after the cloud boom).

Spot prices remain resilient; likely robust exports

Memory spot prices have remained resilient despite stock-market volatility and concerns about an AI bust. DDR5 prices have risen steadily, gaining 5-6% MTD. Although legacy DDR4 and NAND spot prices fell a further 1-2% this week, they remain up 10-20% QoQ. The 4Q DRAM contract prices currently under negotiation appear broadly in line with our forecasts for 5-10% increases or better, led by high-end LPDDR5. However, we see minimal contract price increases for PC/smartphone NAND (similar to softening spot prices), vs mid-single-digit% increases for eSSD. Separately, Korea's semis exports remained strong in Oct (up 200%+ YoY during the first 20-day period).

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Simon Woo, CFA >>
Research Analyst
Merrill Lynch (Seoul)
+82 2 3707 0554
simon.woo@bofa.com

Dai Shen >>
Research Analyst
Merrill Lynch (Hong Kong)
dai.shen@bofa.com

Vivek Arya
Research Analyst
BofAS
vivek.arya@bofa.com

Mikio Hirakawa >>
Research Analyst
BofAS Japan
mikio.hirakawa@bofa.com

Matt Shin >>
Research Analyst
Merrill Lynch (Seoul)
matt.shin2@bofa.com

Exhibit 1: DDR5 slightly up vs softening DDR4 and NAND

Spot-market prices among DRAM and NAND

US$	Current	WoW	QoQ	YoY
DRAM spot				
16Gb DDR5	61.3	1%	21%	416%
16Gb DDR4	82.9	-1%	0%	319%
8Gb DDR4	47.1	2%	12%	466%
4Gb DDR3	13.8	1%	3%	365%
NAND spot				
1Tb wafer	29.7	-2%	17%	363%
512Gb wafer	18.6	-1%	-2%	306%
256Gb wafer	18.4	0%	76%	676%

Source: DRAMeXchange

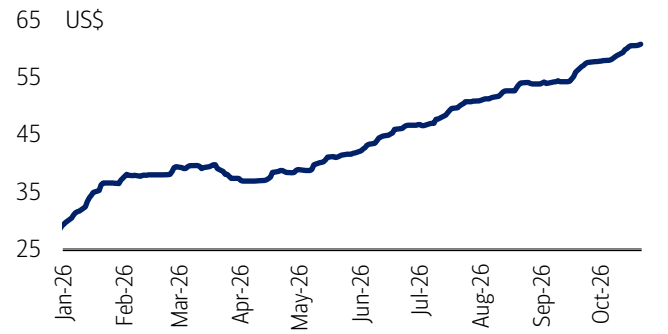
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AI: Artificial intelligence
ASP: Average selling price
DDR4/5: 4th/5th gen double-data rate DRAM
DRAM: Dynamic random-access memory
eSSD: Enterprise solid-state drive
HBM: High bandwidth memory
HBM4/4e: 6th/7th gen of HBM
LPDDR5: Low-power DDR5
LTA: Long-term agreement
NAND: Not-AND memory

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Key 6 memory price charts

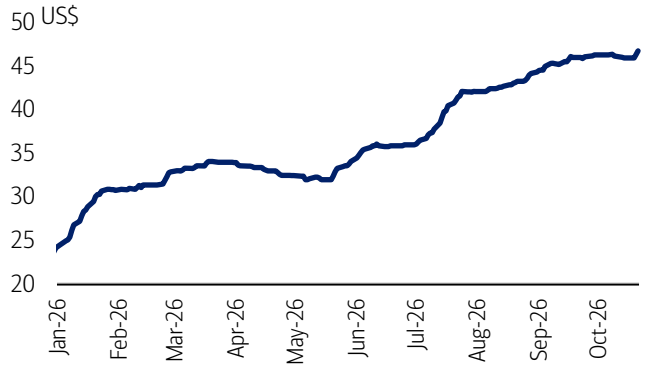
Exhibit 2: DDR5 spot further recovered this week, following last week's sharp rally to hit a record-high $61, with prices now up over 100% YTD and extending the strong upcycle that began in 4Q25
16Gb DDR5 spot price – daily, Jan '26 - Oct '26



Source: DRAMeXchange

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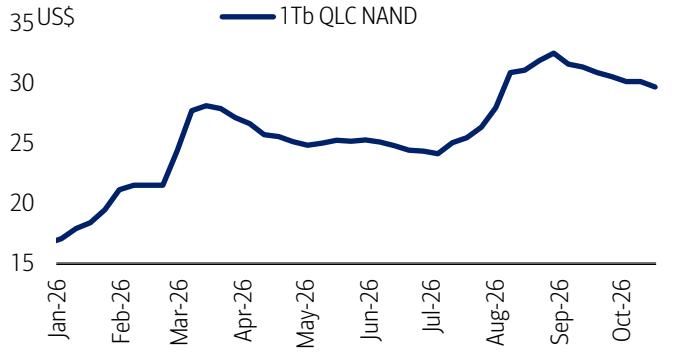
Exhibit 3: 8Gb DDR4 spot prices up this week to hit all-time high of $47, with no signs of a downturn yet
8Gb DDR4 spot price trend, Jan '26 - Oct '26



Source: DRAMeXchange

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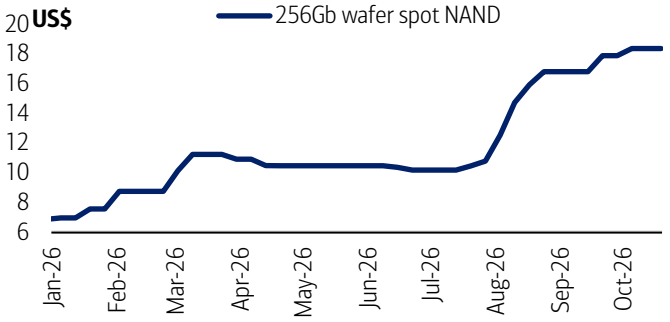
Exhibit 4: NAND spot prices remained weak through Sep and Oct, after recovering through July and Aug
1Tb NAND wafer spot price – weekly, Jan '26 - Oct '26



Source: DRAMeXchange

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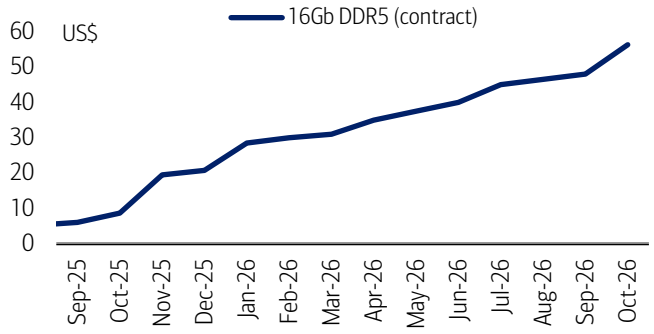
Exhibit 5: In contrast, low-density NAND wafer spot price rebounded sharply through Aug-Sep and early-Oct as well, but remained stable this week
256Gb NAND wafer spot price – weekly, Jan '26 - Oct '26



Source: DRAMeXchange

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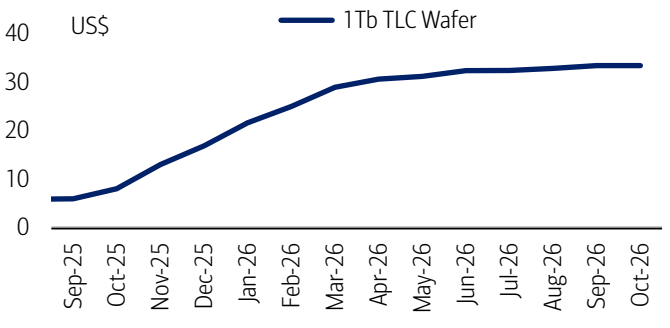
Exhibit 6: TrendForce expects Oct contract price to rise high teen%; no sign for price cut even after 500%+ rally YoY
16Gb DDR5 contract price trend, Sep '25- Sep '26



Oct-26 forecast is TrendForce estimate released on 16th Oct
Source: TrendForce

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Exhibit 7: Oct contract price broadly flat similar to 2Q26-3Q26 trend
1Tb NAND wafer contract price trend, Sep '25- Sep '26



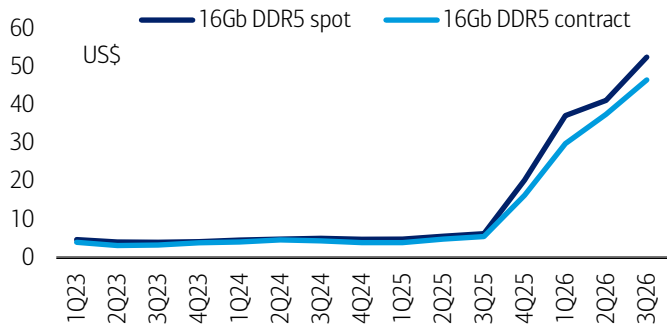
Oct-26 forecast is TrendForce estimate released on 16th Oct
Source: TrendForce

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Quarterly average price trend

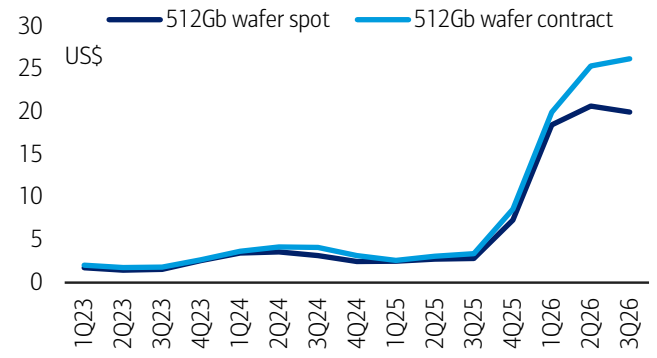
Exhibit 8: DRAM spot and contract prices reached record highs of ~$48–55, with gains moderating in 2Q–3Q26 after the sharp rally seen in 4Q25–1Q26.
DRAM spot and contract price - quarterly average trend



Source: DRAMeXchange, TrendForce

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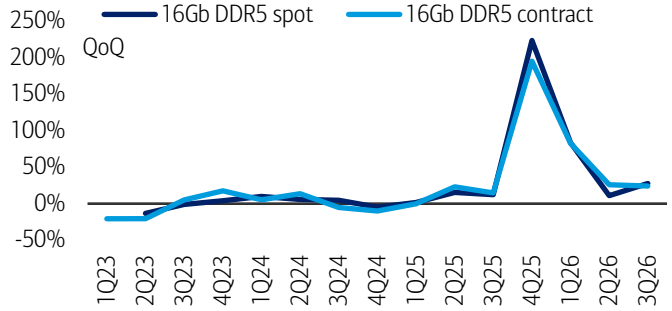
Exhibit 9: NAND wafer contract prices have surged significantly ahead of spot prices, with now at record-high levels; spot slightly corrected
NAND wafer spot and contract price - quarterly average trend



Source: DRAMeXchange, TrendForce

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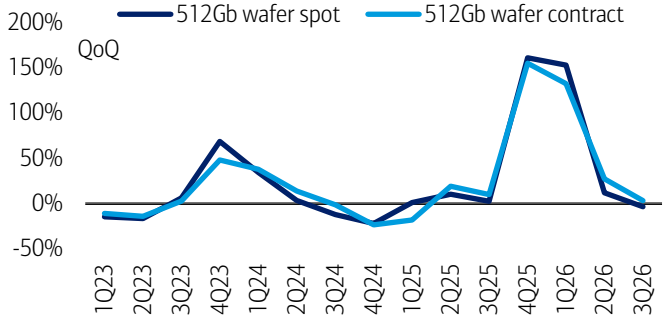
Exhibit 10: Momentum remained positive at +20-30% QoQ in 3Q26, improving from 2Q26 levels but well below the exceptional rally seen in 4Q25–1Q26.
PC DRAM spot / contract prices (quarter-avg) – QoQ change



Source: TrendForce, DRAMeXchange, BofA Global Research

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Exhibit 11: Prices were broadly flat in 3Q26, versus +15-30% QoQ growth in 2Q26, following the sharp ~150% rallies seen in both 4Q25 and 1Q26.
NAND spot / contract prices (quarter-avg) – QoQ change

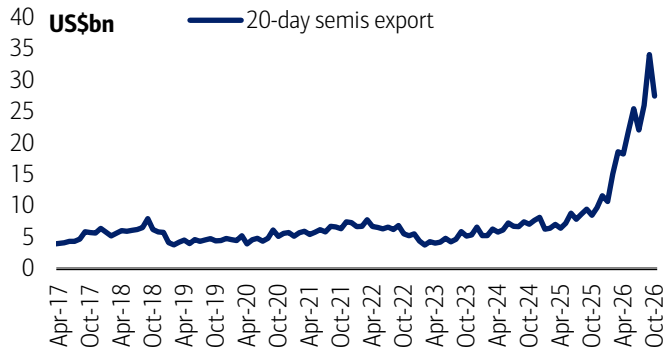


Source: TrendForce, DRAMeXchange, BofA Global Research

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Korea semis exports

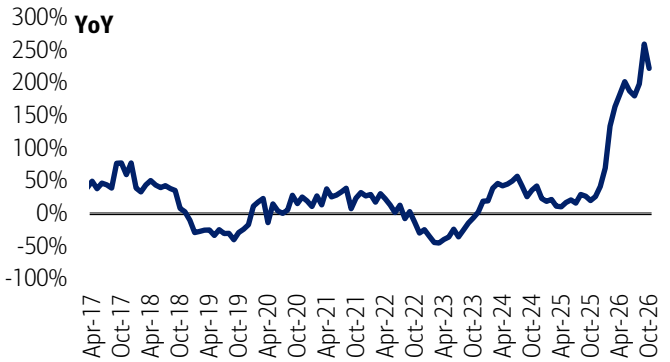
Exhibit 12: Sharp MoM decline in Oct (US$27.5bn; -19% MoM), but still close to record-high and nearly 3.5x higher than 2025-avg (US$8.2bn)
Korea semis exports – first 20 days of month (US$bn)



Source: MoTIR

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Exhibit 13: Robust YoY growth continued in Oct (+222.5%); already nine consecutive months of triple-digit rebound
Korea semis exports – YoY in first 20 days of month



Source: MoTIR

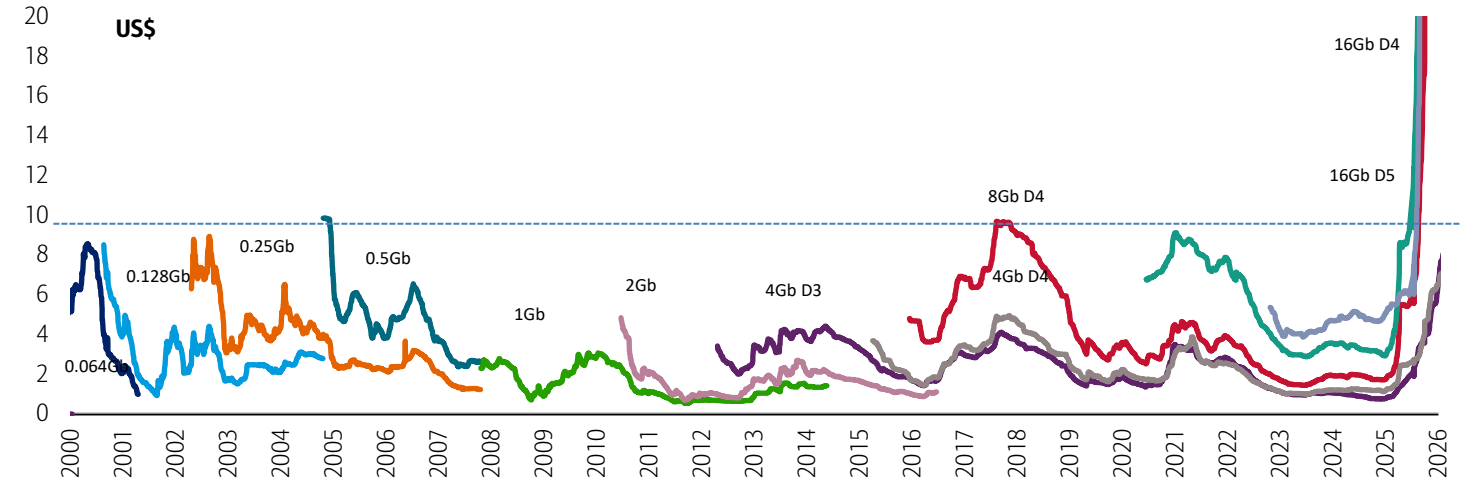
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Memory spot price long term trend

Exhibit 14: DRAM spot prices rebounded gradually through June-Oct, to hit all-time high levels. Prices have reached multi-decade highs, with 16Gb DDR5 (for PC) at ~$60 and DDR4 at ~$82, driven by supply reallocation toward HBM and server DRAM—well above the prior ~$10 peak in Oct-17

DRAM spot price – long-term trend (2000-2026)

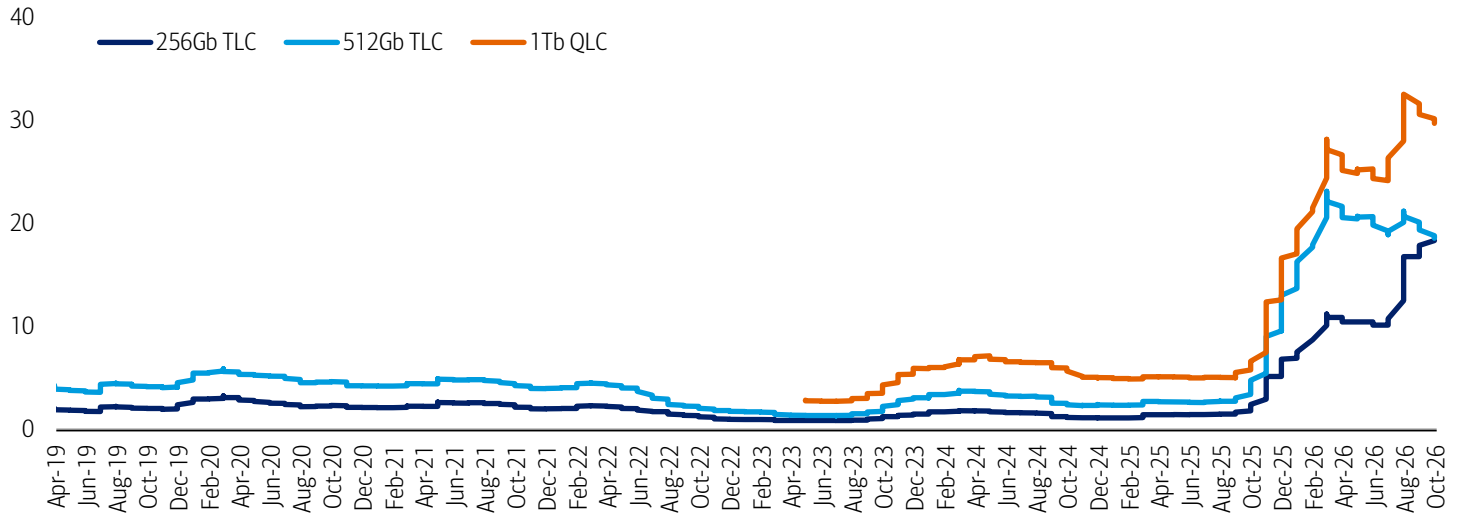


Source: DRAMeXchange

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Exhibit 15: NAND wafer prices remained weak through September and October, extending the largely flat-to-lower trend seen since 2Q26 after the strong 4Q25–1Q26 rally. Meanwhile, 256Gb prices recovered in September and early October before stabilizing this week.

NAND wafer spot price – long-term trend (2019-2026)



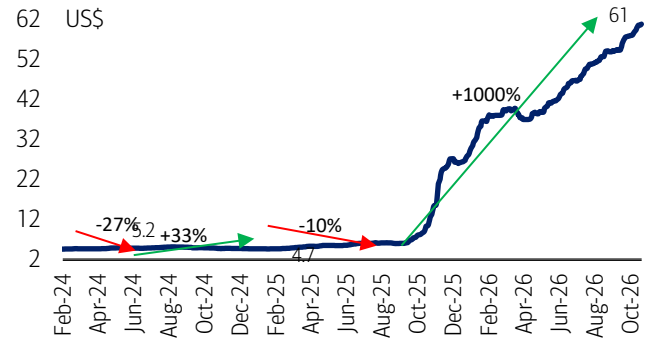
Source: DRAMeXchange

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DRAM spot and contract price trend

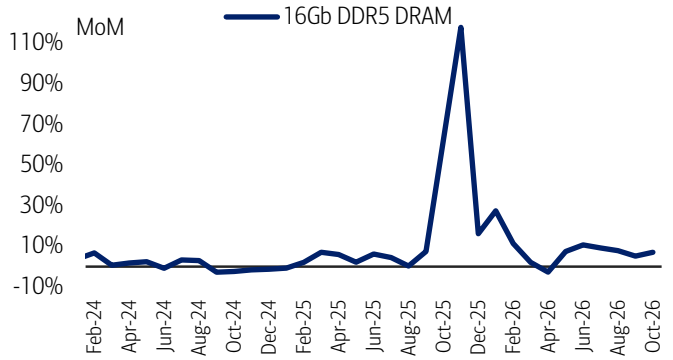
Exhibit 16: Price recovery extended from May to Sep and through Oct, reaching a record high of $61, price strongly recovered in 4Q25, before moderating during Feb–Apr.
16Gb DDR5 spot price – daily, Feb '24 - Oct '26



Source: DRAMeXchange

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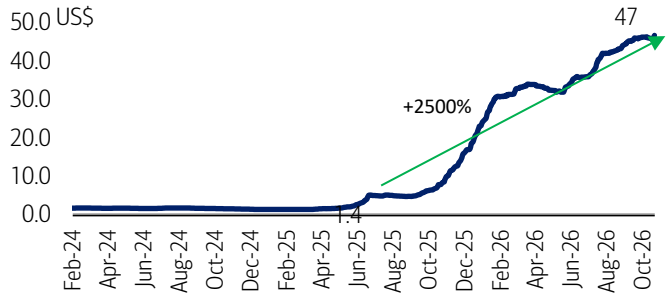
Exhibit 17: Still up 5-7% each in Sep/Oct following a modest uptick observed through May/Jun/Jul/Aug (+7-10%)
16Gb DDR5 spot month-average price – MoM change, Feb '24 - Oct '26



Source: DRAMeXchange

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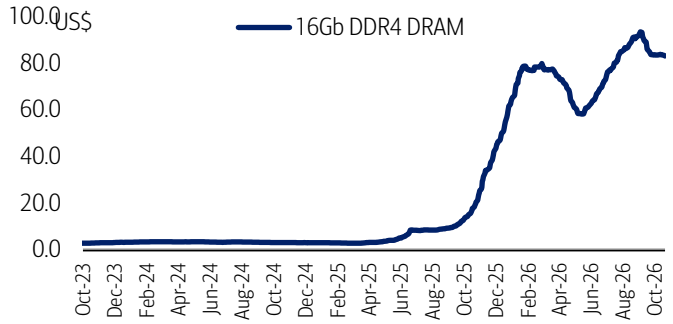
Exhibit 18: 8Gb DDR4 spot prices recovered this week to hit an all-time high of $47; still well above the prior ~$10 peak in October 2017, driven by ongoing supply cuts from major vendors
8Gb DDR4 spot price trend, Feb '24 - Oct '26



Source: DRAMeXchange

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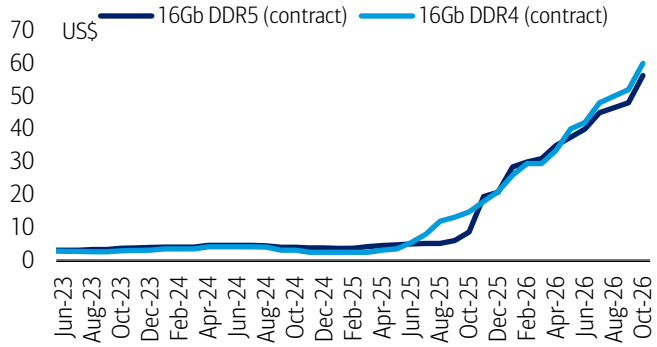
Exhibit 19: Remained flat through Oct, but price down 10% after hitting peak in early-Sep, price still ~2000% above the Oct-25 low (~$3)
16Gb DDR4 spot price trend, Oct '23 - Oct '26



Source: DRAMeXchange

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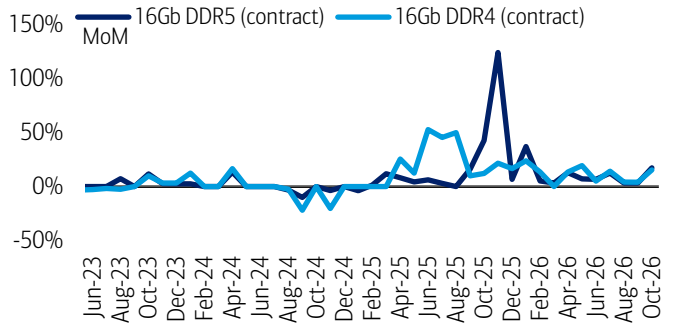
Exhibit 20: DDR4 and DDR5 (16Gb) contract prices are similar at US$50 level – DDR5 price premium no longer exists due to DDR4 shortage
16Gb DDR5 vs 16Gb DDR4 contract price trend, Jun '23- Oct '26



Oct-26 forecast is TrendForce estimate released on 16th Oct
Source: TrendForce

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Exhibit 21: DDR5 and DDR4 contract prices sharply up in Oct vs only low-single recovery in Aug/Sep following +10-15% MoM in July; price strong through 2H25-1H26
16Gb DDR5 vs 16Gb DDR4 contract price change - MoM, Jun '23- Oct '26

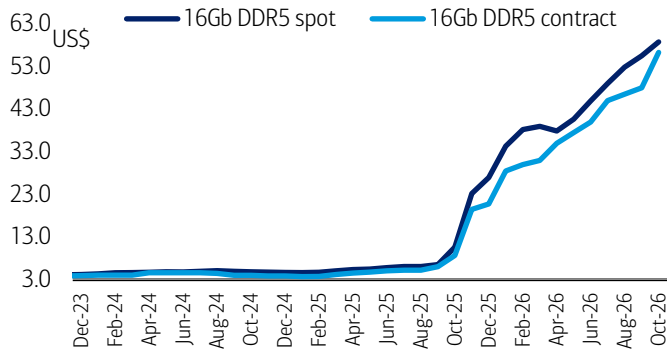


Oct-26 forecast is TrendForce estimate released on 16th Oct
Source: TrendForce

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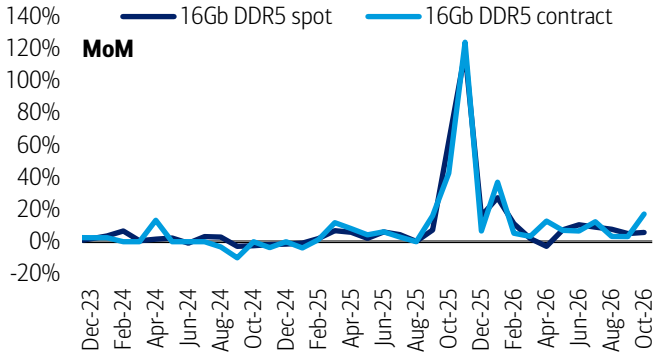
Exhibit 22: 16Gb DDR5 spot and contract prices are in the range of US$55-60 vs historical range of US$3-5
16Gb DDR5 spot vs contract price trend (US$), Dec '23-Oct '26



Oct-26 forecast is TrendForce estimate released on 16th Oct
Source: DRAMeXchange, TrendForce

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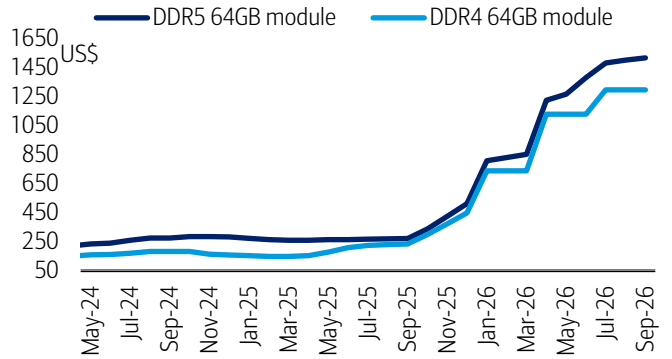
Exhibit 23: Oct contract price recovery stronger (15%+) than spot (6%); prices had witnessed a sharp rally in Oct–Dec'25
16Gb DDR5 spot vs contract price trend – MoM change, Dec '23-Oct '26



Oct-26 forecast is TrendForce estimate released on 16th Oct
Source: DRAMeXchange, TrendForce

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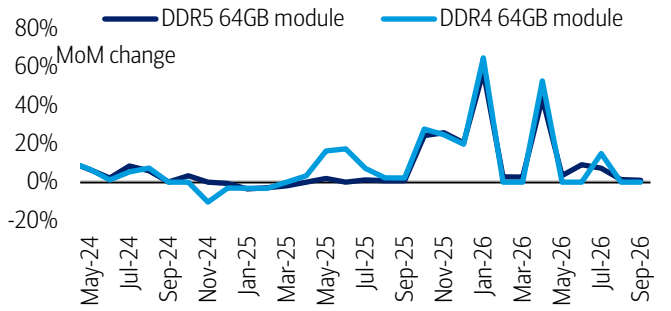
Exhibit 24: Current price of 64GB DDR4/DDR5 modules hit all-time high of more than US$1,000 level (DDR5: US$1,515, DDR4: US$1,295)
Server DRAM contract price trend – DDR5 vs DDR4 modules, May'24-Sep'26



Source: TrendForce

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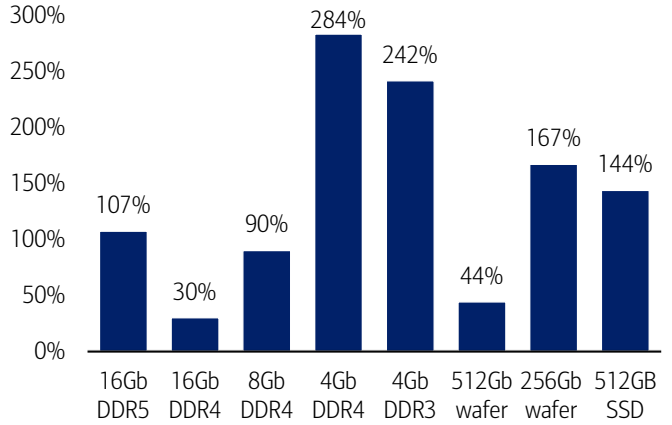
Exhibit 25: DDR5/DDR4 server contract prices remained broadly flat in Aug/Sep after recovering +7-15% MoM in July, remained stable in May/Jun-26
MoM change of server DRAM contract prices, May'24-Sep'26



Source: TrendForce

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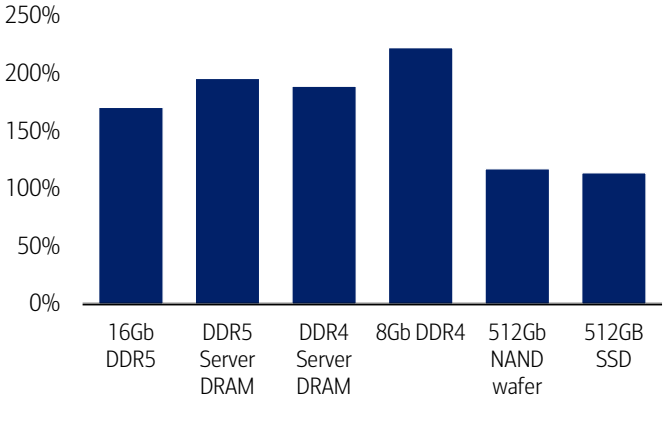
Exhibit 26: DRAM and NAND spot price overall up 100%+ YTD,
YTD change of DRAM and NAND spot price



Source: DRAMeXchange, TrendForce, BofA Global Research

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Exhibit 27: DRAM and NAND contract price overall up 100-200%YTD,
YTD change of DRAM and NAND contract price



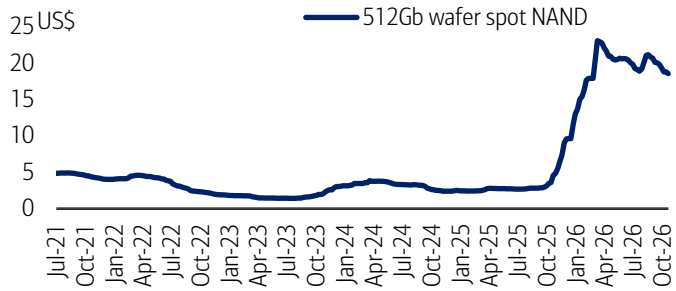
Source: DRAMeXchange, TrendForce, BofA Global Research

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NAND spot and contract price trend

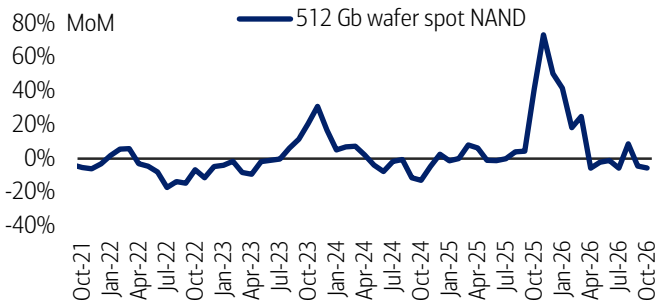
Exhibit 28: NAND spot prices softened in Sept and through-Oct, following a sharp recovery in Aug; remained largely flat to slightly weaker through 2Q26; prices are still up 40-50% YTD and nearly 8x above the February 2025 trough
512Gb NAND wafer spot price – weekly, Jul'21 - Oct'26



Source: DRAMeXchange

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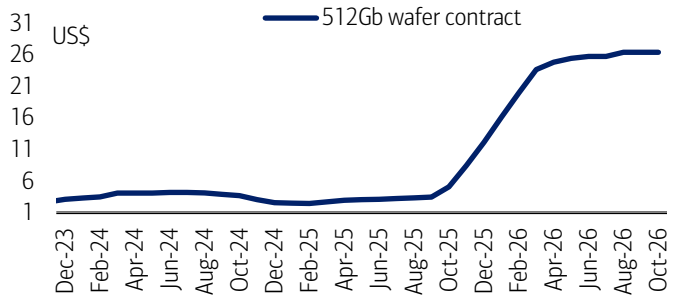
Exhibit 29: Down 5-6% in Sep and Oct following a sharp recovery in Aug (+8%). Remained subdued through Apr-Jul (down 2-5%) following strong 4Q25 (+150%) and 1Q26 (+80%) rally.
512Gb NAND wafer spot average – MoM change, Oct'21 - Oct'26



Source: DRAMeXchange

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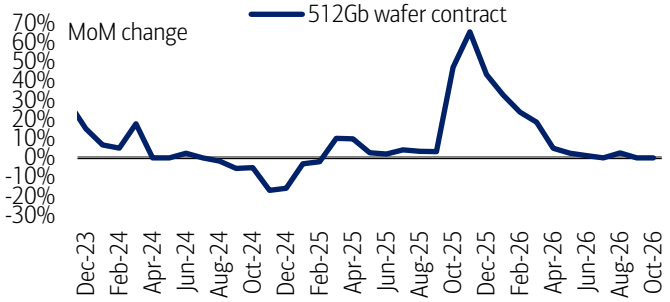
Exhibit 30: Current contract price at ~$27, around 10x vs Feb-25 bottom of $2.5; prices significantly up in 4Q25/1H26 and remained broadly stable thereafter
NAND wafer contract price trend, Dec '23-Oct '26



Oct-26 forecast is TrendForce estimate released on 16th Oct
Source: TrendForce, DRAMeXchange, BofA Global Research

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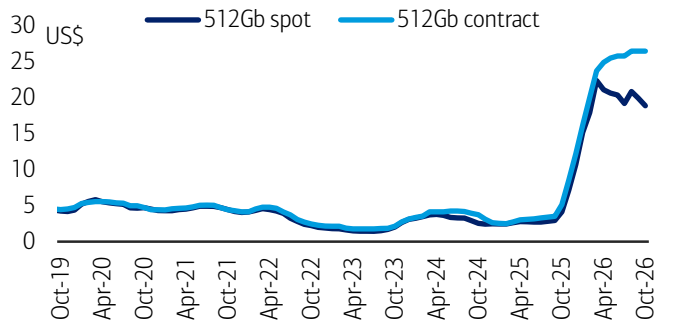
Exhibit 31: Broadly flat or only low-single digit recovery observed through Apr-Oct following Jan/Feb/Mar'26 rally (up 20-30% MoM) and Oct/Nov/Dec upturn (up 40-60% MoM)
MoM change of NAND contract price (512Gb wafer), Dec '23-Oct '26



Oct-26 forecast is TrendForce estimate released on 16th Oct
Source: DRAMeXchange, TrendForce, BofA Global Research

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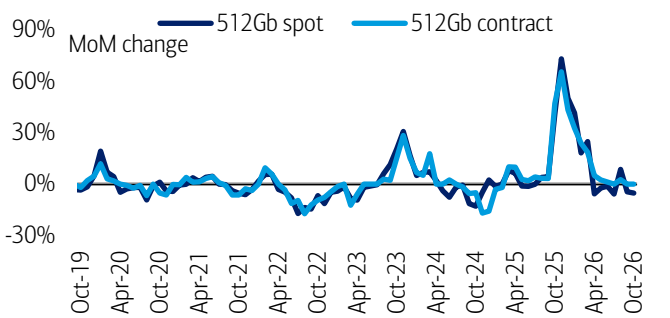
Exhibit 32: Current NAND contract price still near record-high level vs modest correction for spot; that said, the contract prices are still a few times higher than 2025 summer level
512Gb NAND wafer spot vs contract price trend (US$), Oct'19 - Oct'26



Oct-26 forecast is TrendForce estimate released on 16th Oct
Source: DRAMeXchange, TrendForce

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Exhibit 33: NAND contract price flat in Sep/Oct vs spot fell 4-5% MoM post Aug recovery and normalizing in Apr-Jul; strong rally seen in Oct'25 – Dec'25 (up 40-50% per month) and Jan-Mar'26 (20-30% per month)
512Gb NAND spot vs contract price trend – MoM change, Oct'19 - Oct'26



Oct-26 forecast is TrendForce estimate released on 16th Oct
Source: DRAMeXchange, TrendForce

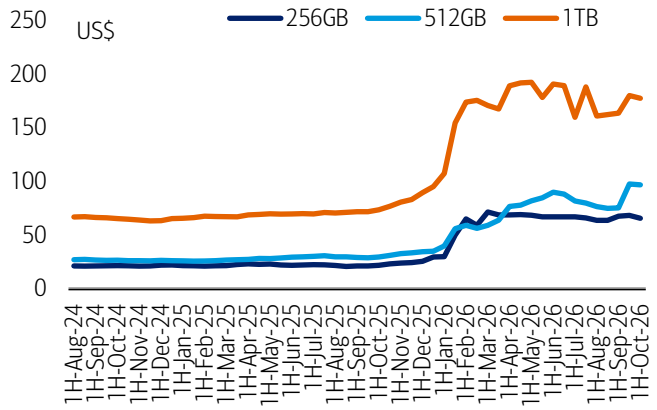
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SSD and panel price trend

Exhibit 34: Oct SSD price slightly corrected or remained stable vs sharp Sep recovery; strong rally was witnessed in 1H26 following 2025 gradual recovery

Client SSD price trend – mostly for PC (not server), Jun'24 - Sep'26

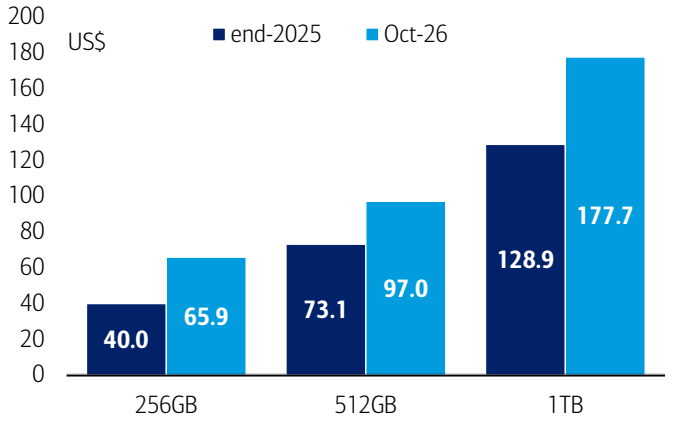


Note: SSD prices are as of 14 Oct 2026, reported by DRAMeXchange. TB = terabyte.
Source: DRAMeXchange

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Exhibit 35: Oct prices have more than doubled compared to end-2025 levels, while the 2025 increase was more modest at around 35–40%

Client SSD (for PC) price comparison – current vs. end-2025

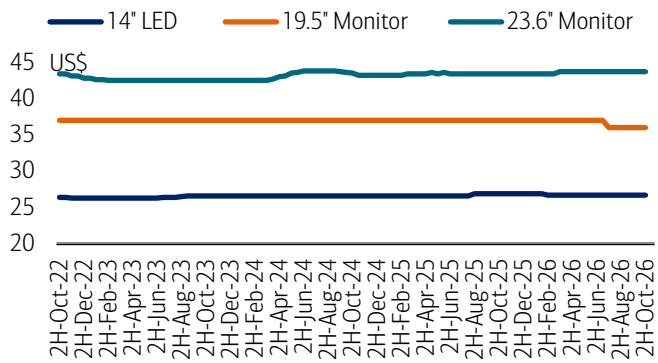


Note: SSD prices are as of 14 Oct 2026, reported by DRAMeXchange
Source: DRAMeXchange

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Exhibit 36: Broadly stable YTD – more China dominant supply vs minimal from Korea

LCD IT panel price trend – US$/panel, Oct '22-Oct '26

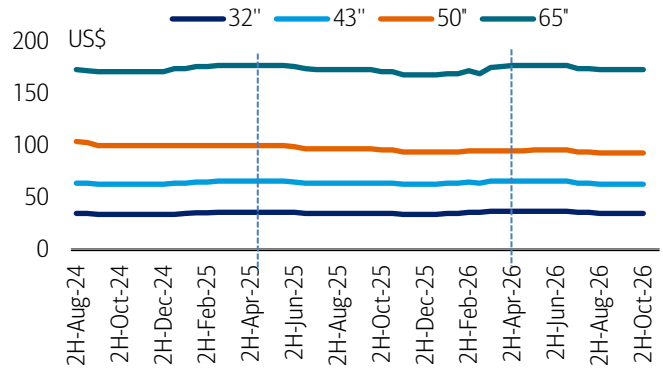


Note: Panel price update from WitsView as of 21 Oct 2026
Source: WitsView

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Exhibit 37: Prices slightly up through 1H26 vs slightly down/stable thereafter; prices remained broadly flat through 2025

LCD TV panel price trend – US$/panel, Aug '24-Oct '26



Note: Panel price update from WitsView as of 21 Oct 2026
Source: WitsView

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Valuation table and YTD price performance

Exhibit 38: Memory stocks rebounded as strong AI/HBM demand and tight industry supply outweighed lingering concerns around Chinese competition. Fundamentals remain firmly intact, supported by robust HBM demand, tight memory supply conditions, and continued expansion in hyperscaler AI CapEx. Despite the recent pullback, most DRAM names continue to trade at relatively undemanding valuations of roughly 3-5x 2026-27 earnings, well below broader AI semiconductor peers

Valuation comparison among memory and semiconductor supply chain stocks

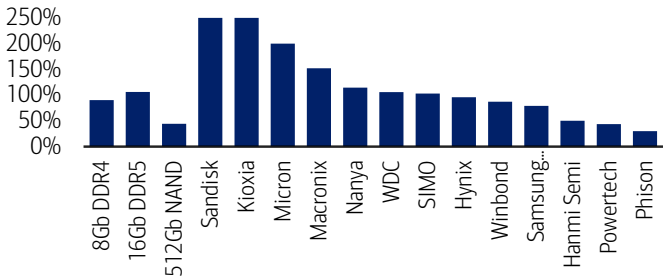
	Ticker	Rating	Price (Local)	Mcap ($bn)	P/E			P/Book			EV/EBITDA		EV/Sales		ROE			Div. yield	
					FY26	FY27E	FY28E	FY26	FY27E	FY28E	FY27E	FY28E	FY27E	FY28E	FY26	FY27E	FY28E	FY27E	FY28E
Memory/DRAM																			
SK Hynix	HXSCF	C-1-7	1,223,000	704.7	3.5	3.4	3.4	2.3	1.5	1.1	2.0	1.9	1.6	1.5	98.2%	52.9%	38.1%	6.2%	7.8%
Samsung	SSNLF	C-1-7	214,500	951.8	5.0	3.7	3.8	2.2	1.5	1.1	2.2	2.3	1.4	1.3	52.4%	46.1%	32.9%	5.0%	5.2%
Micron	MU	C-1-7	856.49	969.1	11.3	5.0	4.3	7.0	3.5	2.4	3.8	3.4	3.2	2.7	89.7%	95.8%	68.8%	0.1%	0.1%
Nanya	NNYAF	B-1-7	414.00	43.8	5.6	4.0	4.9	2.9	1.8	1.5	2.3	2.8	1.8	1.8	75.3%	56.3%	34.2%	10.0%	10.2%
Memory/NAND																			
Kioxia	XIYUF	C-1-9	12,755	133.6	4.0	3.0	2.4	3.6	1.6	1.0	1.8	1.4	1.5	1.2	144.4%	74.2%	50.5%	0.0%	0.0%
Sandisk	SNDK	C-1-9	1,260.93	182.2	n/a	5.4	5.1	11.9	3.4	2.0	3.5	3.3	3.0	2.8	88.0%	103.2%	52.1%	0.0%	0.0%
Western Digital	WDC	C-1-7	354.19	132.6	n/a	14.9	8.9	15.0	10.7	7.1	11.6	7.3	6.5	4.5	54.9%	85.1%	94.7%	0.2%	0.2%
SIMO	SIMO	C-1-7	187.96	6.4	16.1	17.6	14.2	5.2	4.2	3.4	12.3	10.0	3.0	2.6	39.1%	27.2%	27.4%	1.5%	1.7%
Phison	PISNF	C-1-8	1,880.00	12.7	5.2	8.9	13.9	3.1	2.6	2.4	5.6	8.6	1.3	1.5	92.3%	37.0%	20.7%	6.1%	0.0%
US/Taiwan semis																			
NVIDIA	NVDA	C-1-7	212.21	5,114.3	23.3	13.5	9.2	17.7	10.1	6.3	11.5	8.1	7.7	5.5	96.8%	90.7%	77.6%	0.5%	0.7%
AMD	AMD	C-1-9	507.93	829.2	66.7	31.2	20.6	11.1	8.1	5.7	24.4	16.1	9.1	6.3	18.3%	31.0%	34.8%	0.0%	0.0%
Broadcom	AVGO	C-1-7	323.24	1,543.0	27.9	17.4	10.4	14.1	8.9	5.3	13.6	8.1	9.1	5.4	60.0%	65.1%	66.3%	0.9%	1.0%
Intel	INTC	C-1-9	85.49	451.8	58.2	42.5	29.3	3.4	3.2	2.9	14.8	11.6	5.9	5.1	5.9%	7.9%	10.6%	0.0%	0.0%
Arm	ARM	C-2-9	210.10	224.4	94.6	71.5	54.0	20.7	15.3	11.2	54.0	40.3	26.8	20.7	25.1%	25.0%	24.5%	0.0%	0.0%
Qualcomm	QCOM	C-3-7	172.83	184.6	16.4	16.3	13.9	6.3	5.9	5.1	13.2	11.5	4.3	3.8	44.4%	36.9%	38.0%	2.1%	2.1%
TSMC	TSMWF	B-1-7	2,385	1,896.4	21.3	15.0	11.9	8.0	5.5	3.9	10.0	8.1	7.5	6.1	44.0%	43.3%	38.7%	1.0%	1.0%
MediaTek	MDTKF	C-1-7	4,010	197.3	58.5	24.4	10.1	14.8	9.9	5.5	19.3	8.2	4.8	2.5	26.1%	48.1%	69.3%	3.3%	7.9%
Equipment																			
ASML	ASMLF	B-1-7	1,419.60	617.0	32.5	23.5	16.4	21.1	15.4	10.4	18.8	13.4	9.2	7.0	73.9%	75.1%	75.3%	1.1%	1.4%
AMAT	AMAT	C-1-7	375.58	298.1	29.4	20.3	16.0	10.6	7.4	5.3	17.3	13.8	6.4	5.3	41.9%	42.9%	38.2%	0.6%	0.7%
Lam	LRCX	C-1-7	230.15	287.5	39.5	24.4	18.9	23.1	14.6	10.1	19.8	16.1	8.3	6.7	65.7%	73.5%	62.6%	0.5%	0.6%
TEL	TOELF	C-1-7	9,458	140.8	27.5	19.9	15.7	8.3	6.7	5.4	13.8	11.1	5.2	4.2	33.6%	37.3%	38.1%	2.5%	3.2%
Hanmi Semi	HNSIF	C-1-7	191,200	13.8	56.5	31.6	18.0	19.4	12.9	8.2	23.4	13.2	12.8	7.7	39.4%	48.9%	55.7%	1.1%	1.6%
BESI	BESVF	C-2-7	157.25	14.3	35.8	23.8	21.1	17.0	13.8	13.3	18.6	16.3	9.2	8.1	61.0%	63.8%	64.1%	4.4%	6.2%
ASMPT	ASMVF	C-1-7	123.10	6.6	29.1	18.6	13.1	2.9	2.6	2.3	11.3	8.3	1.8	1.3	10.1%	14.7%	18.7%	2.2%	2.6%

Source: Company reports, BofA Global Research estimates; Mcap = market capitalization, ROE = return on equity, Div = dividend. Price as of latest close.

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Exhibit 39: Memory stocks rebounded this week, following last week's China threat driven sell-off, supported by renewed confidence in strong AI/HBM demand and tight memory supply, share prices remain well above end-2025 levels.

Memory companies – 2026 YTD stock performance comparison



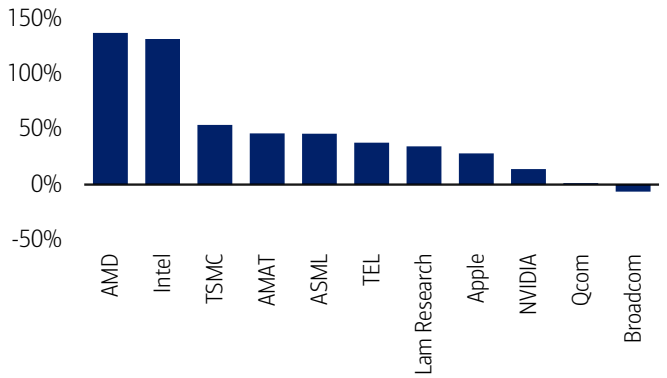
*SanDisk and Kioxia shows above 250% YTD return

Source: Bloomberg, DRAMeXchange

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Exhibit 40: CPU names such as Intel and AMD outperforming other US big tech names such as NVIDIA/Apple/QCOM

Global major tech companies – 2026 YTD stock performance comparison



Source: Bloomberg

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